Loop structure between LLM and Deterministic Core to cut inference cost.
Feed it a lesson plan and it already knows what to 'listen' for rather than solving the hard problem, over and over.

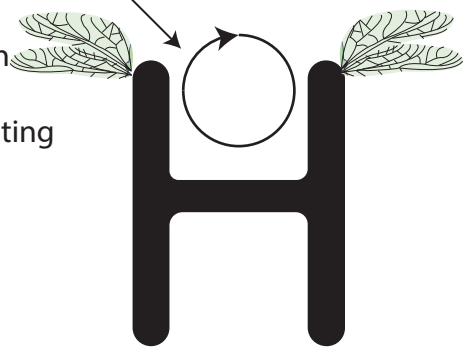
The "neuro" part of neurosymbolic AI

Multi-modal inputs
Speech
Students' written reflections
for semantic routing

Open-Source LLM

Initially powered by API calls to REALLMs
Ideally, locally run on cheap hardware using platforms like Gemma 4

Mixture of Experts



Multi-modal outputs
(text, number lines, speech for accessibility)
Discussion questions
Assessing questions
Advancing questions
Knowledge-graphs
Student pairing suggestions

Deterministic "skeleton" - No 'slop'; 2 always equals 2.
The symbolic 'core' of neurosymbolic AI
Initially, CCSM and State Standards
Automata for student invented strategies
Misconceptions documented in the math education literature
Prolog for 'good ol' fashioned AI', basic pattern recognition
Incoherence frames (Brandom, 2008)
as scene-level architectures (Carspecken, 1999).
Ideally, theorem proving packages for demonstrating why
some student invented strategies lead to contradictions and why
some are 'the same'.